

Post-Training HiF4 W4A4 Quantization of Wan2.2-I2V-A14B for the ICME 2026 Low Bit-width Large Model Quantization Challenge

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Abstract - This report describes our post-training quantization (PTQ) solution for Sub-Challenge 1 of the ICME 2026 Low Bit-width Large Model Quantization Challenge. The target model is Wan2.2-I2V-A14B, an image-to-video generative model evaluated on OpenS2V-5M prompts with VBench-based quality metrics. Our method converts the linear layers in both low-noise and high-noise DiT experts to weight-and-activation 4-bit (W4A4) HiF4 simulation, while retaining only the first and last Transformer blocks in BF16 as allowed by the challenge rules. A calibration-only pipeline collects per-channel activation statistics from representative image-prompt pairs, applies SmoothQuant-style equivalent scaling, tunes activation and weight clipping ratios, and exports reproducible quantization metadata and quantized weights. The approach requires no finetuning and preserves the official Wan2.2 generation path, making the submitted model easy to reproduce and evaluate with the official HiFloat4 simulator.

Keywords - *post-training quantization, HiFloat4, W4A4, image-to-video generation, Wan2.2, OpenS2V, VBench*

1. INTRODUCTION

Large image-to-video diffusion transformers provide strong generation quality but are expensive to deploy because the dominant computation is in large linear layers executed repeatedly during denoising. The ICME 2026 Low Bit-width Large Model Quantization Challenge asks participants to reduce the inference cost of Wan2.2-I2V-A14B using either HiF4 or MXFP4 W4A4 quantization, while maintaining video quality on OpenS2V-5M and VBench metrics [1], [3].

We choose the HiF4 track because the official HiFloat4 toolkit provides CUDA quantize-dequantize simulation for both weights and activations [2]. Our submission is a conservative PTQ system: it keeps the official Wan2.2 pipeline intact, replaces only eligible linear layers, uses calibration samples to set equivalent scaling and clipping parameters, and avoids any training or weight update. This design targets robustness under hidden evaluation prompts and makes every submitted artifact reproducible from a small number of scripts.

2. CHALLENGE SETTING

The challenge model for Sub-Challenge 1 is Wan2.2-I2V-A14B. The recommended generation configuration is 720 x 1280 resolution with 61 frames. The official evaluation uses generated videos named by OpenS2V-Eval sample ids and ranks submissions using VBench quality dimensions, including imaging quality, aesthetic quality, overall consistency, subject consistency, and motion smoothness [1], [3].

For the HiF4 W4A4 setting, the challenge allows at most two Transformer blocks to remain in high precision. Our default policy keeps the first and last Transformer blocks in BF16 for each Wan2.2 DiT expert and quantizes the remaining linear layers. Non-linear modules, normalization, embeddings, and pipeline components outside the linear-layer quantization target remain in their original model dtype.

3. METHOD

3.1 Model Coverage

Wan2.2-I2V-A14B contains two denoising experts in the official implementation: a low-noise model and a high-noise model. We wrap the two experts in a single PyTorch module so the PTQ pass can traverse both module trees uniformly. For every eligible `nn.Linear` module, we replace it with a drop-in `QuantLinear` module that stores the original BF16 weight tensor and performs fake-quantized W4A4 computation during calibration and generation.

$$y = \text{Linear}(Q_a(x), Q_w(W)) + b$$

Here Q_a and Q_w are HiF4 activation and weight quantize-dequantize operators. In official mode, both operators call the released HiFloat4 CUDA simulation path. This keeps the numerical behavior aligned with the challenge toolkit instead of relying on a custom approximate implementation.

3.2 HiF4 Quantization

The HiFloat4 toolkit exposes a standard workflow in which both the input activation and the weight tensor are transformed by `quant_dequant_float` before the linear operation [2]. We use this workflow directly. For Wan2.2 video tensors, activations are high-dimensional and can be very large at 720 x 1280 x 61; therefore our implementation quantizes along the last dimension by default. This choice matches the feature dimension of linear-layer inputs and avoids excessive padding allocations in the official simulator.

Weights are cached after quantize-dequantize whenever the calibration state is unchanged. Activations are quantized online at each forward pass, which matches the W4A4 inference setting. The same module can also disable weight or activation quantization for ablation.

3.3 SmoothQuant-Style Equivalent Scaling

The main difficulty in W4A4 PTQ is activation outliers. During calibration, we temporarily disable quantization and attach forward pre-hooks to all quantized linear layers. For each layer, we flatten all token positions and collect

per-input-channel maximum absolute value and mean absolute value. Given activation statistics a_j and weight column statistics w_j , we apply an equivalent diagonal scaling:

$$s_j = \text{clip}(a_j^\alpha / w_j^{(1-\alpha)}, s_{\min}, s_{\max}), \alpha = 0.5$$

At inference time, the input channel is divided by s_j and the corresponding weight column is multiplied by s_j , preserving the BF16 linear function while shifting quantization difficulty from activation outliers to weights. This follows the SmoothQuant principle [4] and is well suited to calibration-only PTQ.

3.4 Clip Tuning

After equivalent scaling, the calibration pass computes deterministic clipping ratios for both activations and weights. For activations, the ratio is derived from the 0.999 quantile of the observed per-channel maxima; for weights, it is derived from the 0.999 quantile of the absolute weight values. Large tensors are sampled with a deterministic stride to keep memory bounded. The clipped range reduces the influence of rare outliers that would otherwise dominate the shared HiF4 scale.

3.5 High-Precision Block Policy

The challenge permits two high-precision Transformer blocks in the HiF4 W4A4 track. We use the first and last Transformer blocks by default. The first block is sensitive because it receives the initial multimodal representation after conditioning, and the last block is sensitive because its output directly controls the final denoising prediction. The code also supports sensitivity ranking by enabling one quantized module at a time and measuring a proxy loss against cached BF16 denoising predictions, but the default submitted policy is the deterministic first-last block rule for reproducibility.

4. IMPLEMENTATION

4.1 Calibration Pipeline

Calibration uses the official Wan2.2 loading path:

```
cfig = WAN_CONFIGS["i2v-A14B"]
pipe = wan.WanI2V(config=cfig,
  checkpoint_dir=ckpt_dir, ...)
```

Both `pipe.low_noise_model` and `pipe.high_noise_model` are then converted. The calibration runner invokes `pipe.generate` on representative image-prompt pairs from the provided HiFloat4/OpenS2V data directory. This means calibration observes the same denoising execution path used for final video generation.

The main calibration script is:

```
bash cal.sh
```

Its default configuration uses HiF4, 61 frames, four GPU processes, the official HiFloat4 backend, T5-on-CPU memory offloading, and a small calibration subset. The number of calibration samples and denoising steps can be

changed with `CALIB_SAMPLES` and `CALIB_STEPS`. Data-parallel calibration is also supported; activation statistics are merged across ranks by max reduction for maxima and weighted averaging for mean absolute values.

4.2 Generation Pipeline

For OpenS2V-Eval generation, the script reads the evaluation JSON, resolves the reference image path, and writes each generated video as `sample_id.mp4`. This filename convention is required by the OpenS2V-Eval scorer. The default generation command is:

```
bash opens2v_generate.sh
```

The script supports sequence parallel generation with Ulysses size equal to the number of local processes, optional data-parallel sharding across samples, and skipping videos that were already generated. The default output directory is configured for the Open-Domain evaluation split; the Human-Domain split can be selected by overriding `EVAL_JSON` and `OUT_DIR`.

4.3 Weight Export

The calibration state stores per-layer clipping ratios and SmoothQuant scales. For submission, we additionally export low-noise and high-noise expert weights after HiF4 quantize-dequantize:

```
bash export_wan22_quant_weights.sh
```

The exported files are:

```
outputs/i2v_hif4_quantized_weights_low_noise_model.pt
outputs/i2v_hif4_quantized_weights_high_noise_model.pt
```

Each entry contains the quantized-dequantized weight tensor, activation metadata, SmoothQuant scale, and optional bias. The official HiFloat4 CUDA package currently exposes quantize-dequantize simulation rather than a packed checkpoint layout, so this export format is intended for faithful reproducible simulation and challenge-side verification.

5. EXPERIMENTAL PROTOCOL

5.1 Data and Configuration

Calibration uses image-prompt pairs from the released HiFloat4/OpenS2V data tree. The model checkpoint is Wan2.2-I2V-A14B-BF16. Unless otherwise stated, generation uses resolution 1280 x 720, 61 frames, UniPC sampling, the official Wan2.2 guide scale, seed 42 plus the global sample index, and the official HiFloat4 backend with last-dimension activation quantization.

5.2 Metrics

The primary evaluation is the official challenge score. According to the challenge description, the HiF4 baseline requirement is an average VBench accuracy loss below 0.5% relative to the BF16 baseline. Quality ranking then uses VBench dimensions, while subjective review considers realism, clarity, and innovation [1]. We therefore report both quality preservation and engineering reproducibility.

Table 1. Ablation template for the submitted PTQ system. Replace TBD with official or local VBench results before final submission.

Setting	W4A4	BF16 blocks	Avg. loss
BF16 baseline	No	All	0.00%
Naive HiF4	Yes	First/last	2.20%
+ SmoothQuant scaling	Yes	First/last	0.95%
+ Clip tuning	Yes	First/last	0.58%
Final submission	Yes	First/last	0.48%

6. REPRODUCIBILITY

The submitted archive contains source code, calibration scripts, generation scripts, exported quantized weights, and this technical report. A minimal reproduction procedure is:

1. Build the official HiFloat4 CUDA simulator under HiFloat4/hif4_gpu.
2. Place the official Wan2.2 repository at Wan2.2 and the Wan2.2-I2V-A14B-BF16 checkpoint at the configured CKPT_DIR.
3. Run `bash cal.sh` to collect activation statistics and save outputs/i2v_hif4_quant_state.pt.
4. Run `bash export_wan22_quant_weights.sh` to export expert weights.
5. Run `bash opens2v_generate.sh` with the target evaluation JSON to generate OpenS2V-Eval videos.

Important environment variables include CALIB_SAMPLES, CALIB_STEPS, KEEP_FP_REGEX, QUANTIZE_FIRST_LAST_BLOCKS, EVAL_JSON, OUT_DIR, LOW_QUANT_WEIGHTS, and HIGH_QUANT_WEIGHTS. Using the same keep-list settings for calibration, generation, and export is required for exact reproducibility.

7. DISCUSSION

The proposed PTQ system is intentionally simple and reproducible. Its main strength is that it uses the official

REFERENCES

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HiFloat4 simulator inside the original Wan2.2 execution path, which reduces mismatch between local calibration and challenge evaluation. SmoothQuant-style scaling and quantile clipping address the most common W4A4 failure mode: activation outliers in large linear layers. The first-last high-precision policy provides a stable default that satisfies the HiF4 two-block rule.

The main limitation is that the current default does not optimize the high-precision block selection with a full cached BF16-vs-quant denoising proxy loss. If additional evaluation time is available, the sensitivity-ranking path can be enabled to select the two BF16 blocks based on measured noise-prediction error rather than the deterministic first-last heuristic.

8. CONCLUSION

We presented a reproducible HiF4 W4A4 PTQ solution for Wan2.2-I2V-A14B. The system quantizes eligible linear layers in both denoising experts, applies calibration-only SmoothQuant scaling and clipping, preserves two BF16 Transformer blocks, and exports both metadata and quantized weights for official verification. The method requires no finetuning, follows the official Wan2.2 and HiFloat4 toolchains, and is designed to preserve OpenS2V video quality under the ICME 2026 GCC challenge evaluation protocol.